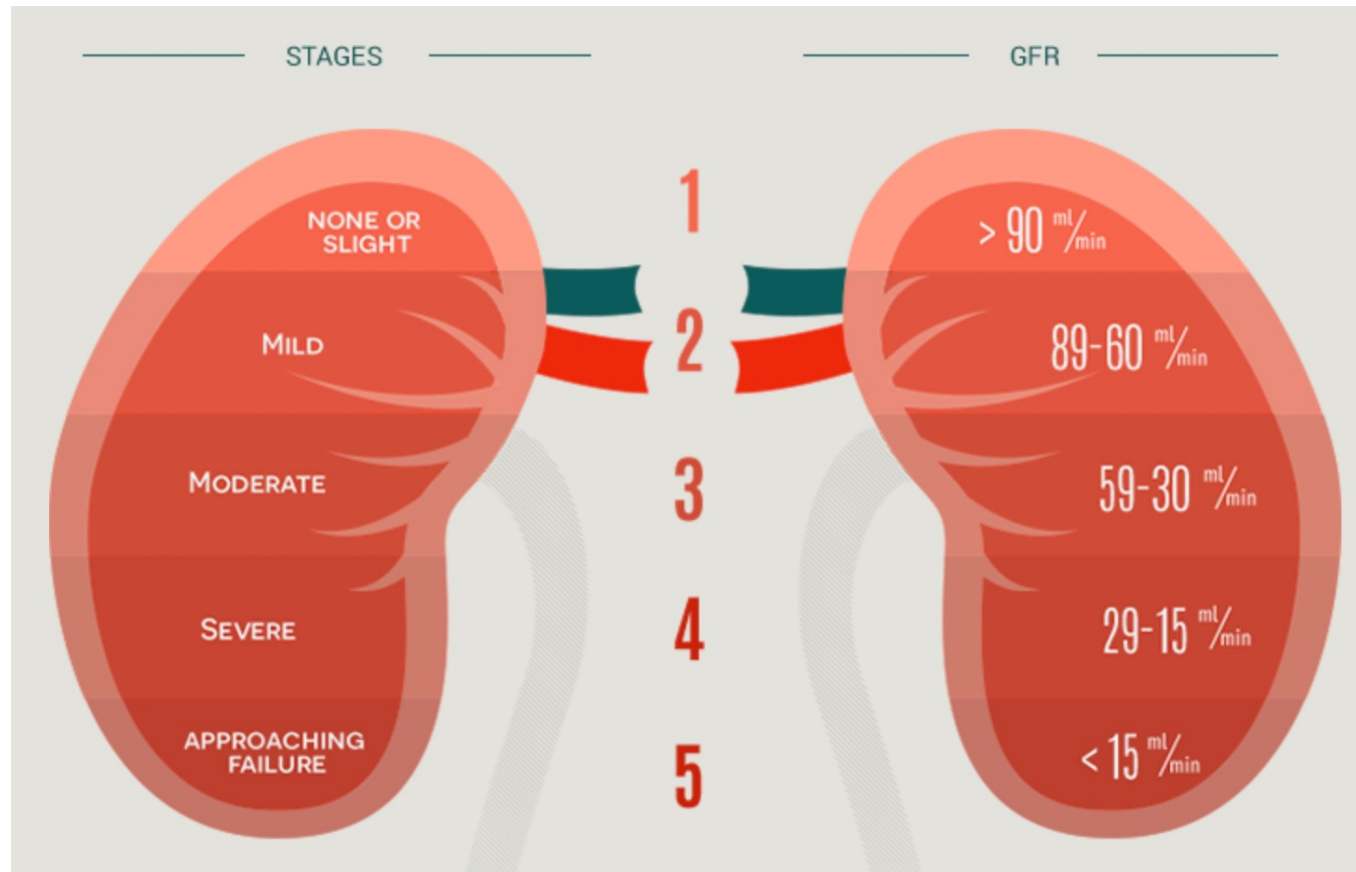


Calibration and Class Balance

Matt Engelhard

Last Time:
Performance metrics are tied to the use case



11%

Probability of Stage IV
CKD within two years

Report Back:

- How might you use this number in clinical practice (be specific)?
- What performance metrics are relevant to this way of using the prediction?

Today

- Consider definitions of algorithmic fairness
- Assessing model calibration
- Class balancing and effects on calibration
 - Discussion
 - Small-group activity
- Post-hoc calibration techniques
- What is fair? Do we emphasize calibration, or discrimination?

Growing concerns about algorithmic bias

You've been using the model for a year now, and you've noticed something potentially concerning.

The model seems to preferentially refer White patients over Black patients. Specifically, you notice that Black patients are referred only if their clinical presentation is more severe compared to White patients who are referred. You are not sure why this is happening.

Report Back: What are some possible sources of this bias, and what could you do to investigate?

Sources of Algorithmic Bias

Many possible sources, but in most cases, it's:

1. Biased outcomes in training data
2. Poor representation of demographics

JOURNAL ARTICLE

Observability and its impact on differential bias for clinical prediction models [Get access >](#)

Mengying Yan, Michael J Pencina, L Ebony Boulware, Benjamin A Goldstein ✉

Journal of the American Medical Informatics Association, Volume 29, Issue 5, May 2022,
Pages 937–943, <https://doi.org/10.1093/jamia/ocac019>

Published: 25 February 2022 **Article history** ▼

What does it mean for a model to be *fair*?

Consider whether predictions are fair between two groups A and B.

One possible solution is *fairness through unawareness*, also called *blindness*, wherein the model is not given information about group status.

Report back:

- What could go wrong if the model is *aware* of group status?
- What could go wrong if the model is *unaware* of group status?

Then, propose an alternative approach to achieve or evaluate fairness.

A troubled calculus

Researchers use race to build disease risk assessment tools. Can removing it help resolve medicine's race crisis?

JYOTI MADHUSOODANAN [Authors Info & Affiliations](#)

SCIENCE • 23 Jul 2021 • Vol 373, Issue 6553 • pp. 380-383 • DOI: 10.1126/science.373.6553.380

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PERSPECTIVE

**Health
Economics****WILEY**

Patient-centered appraisal of race-free clinical risk assessment

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Abstract

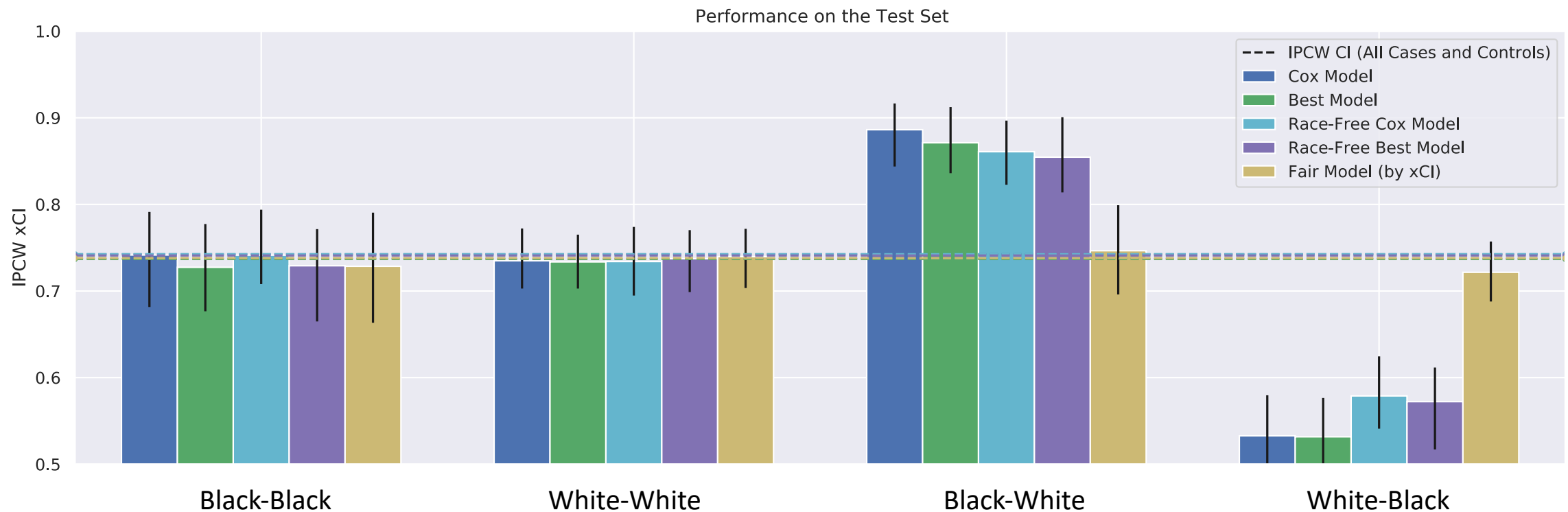
Until recently, there has been a consensus that clinicians seeking to assess patient risks of illness should condition risk assessments on all observed patient covariates with predictive power. The broad idea is that knowing more about patients enables more accurate predictions of their health risks and, hence, better clinical decisions. This consensus has recently unraveled with respect to a specific covariate, namely race. There have been increasing calls for race-free risk assessment, arguing that using race to predict health risks contributes to racial disparities and inequities in health care. In some medical fields, leading institutions have recommended race-free risk assessment. An important open question is how race-free risk assessment would affect the quality of clinical decisions. Considering the matter from the patient-centered perspective of medical economics yields a disturbing conclusion: Race-free risk assessment would harm patients of all races.

KEYWORDS

predicting illness risk, racial health disparities, risk assessment algorithms

Blindness can be ineffective or even harmful.

- In some cases it is effective.
- In other cases it isn't enough to fix the problem.
- In still other cases, it can make the problem worse.



Possible definitions of algorithmic fairness

Consider whether predictions are fair between two groups A and B.

1. Blindness: The model *cannot see* group status
2. Independence: Model predictions are *independent* of group status
3. Separation: Model *performance* is similar in all groups
4. Individual fairness: The model makes *similar predictions* for similar individuals regardless of group status

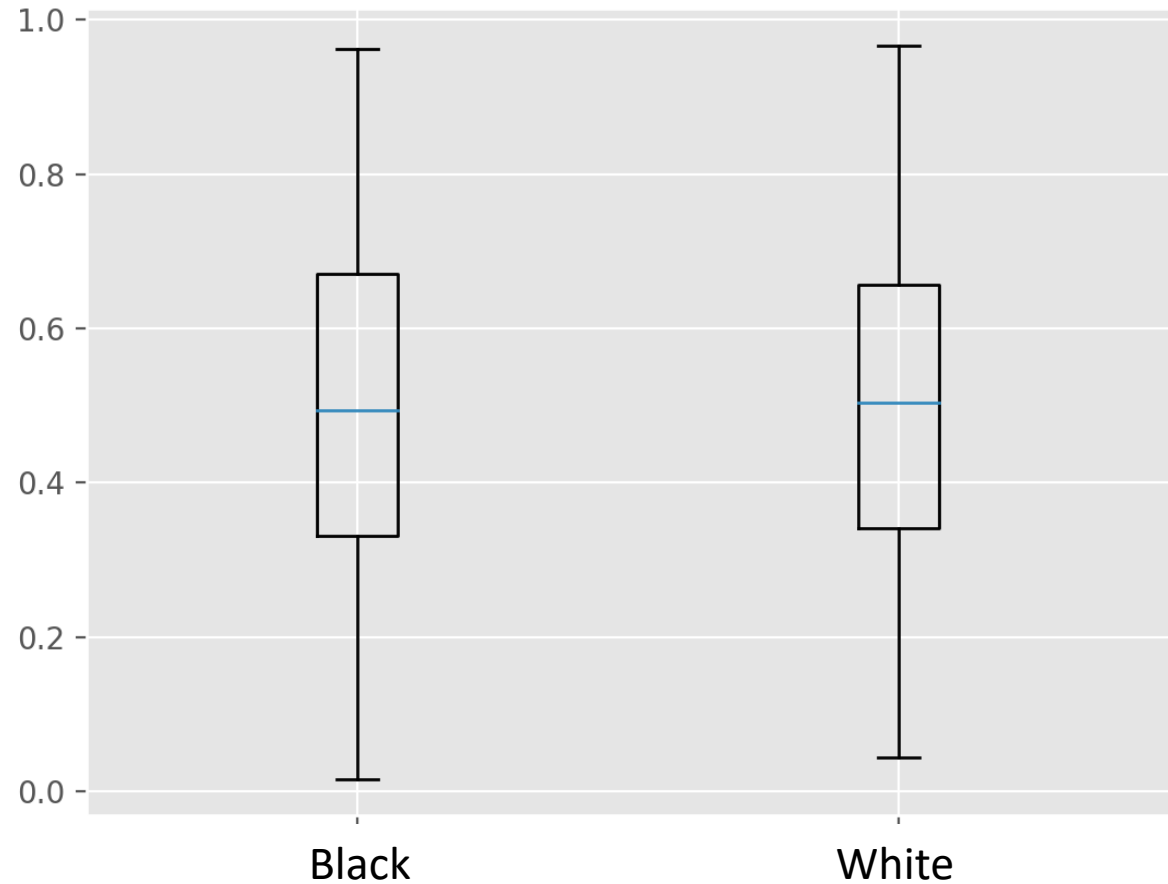
Except under special (& not realistic) circumstances, (1) – (4) are mutually exclusive.

Zooming in on *Independence*

Model predictions are independent of group status.

This implies that the distribution of model predictions is the same in each group.

Is this **fair**?

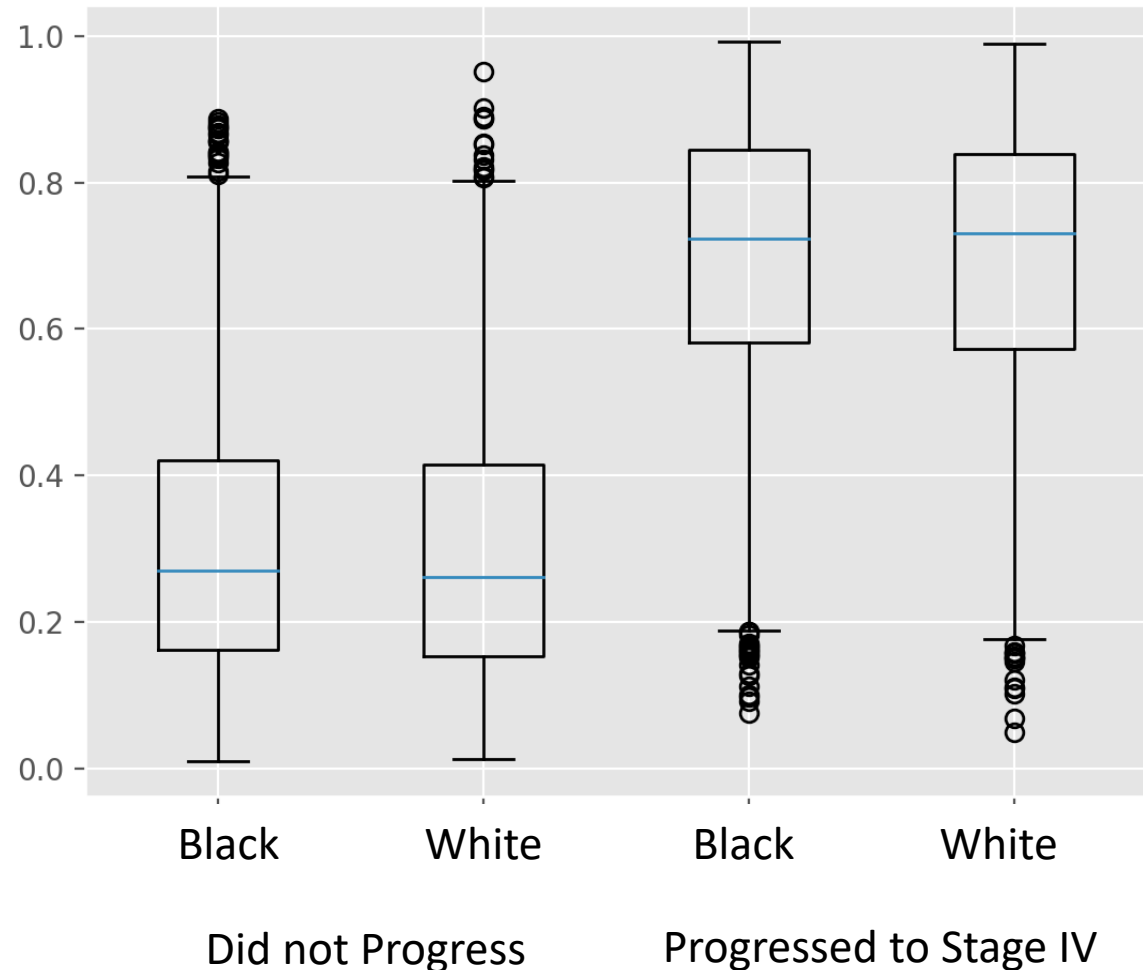


Zooming in on *Separation*

Model predictions are independent of group status *after conditioning on the outcome*.

This implies when considering individuals *with the same outcome*, the distribution of model predictions is the same in each group.

Is this **fair**?



Possible definitions of algorithmic fairness

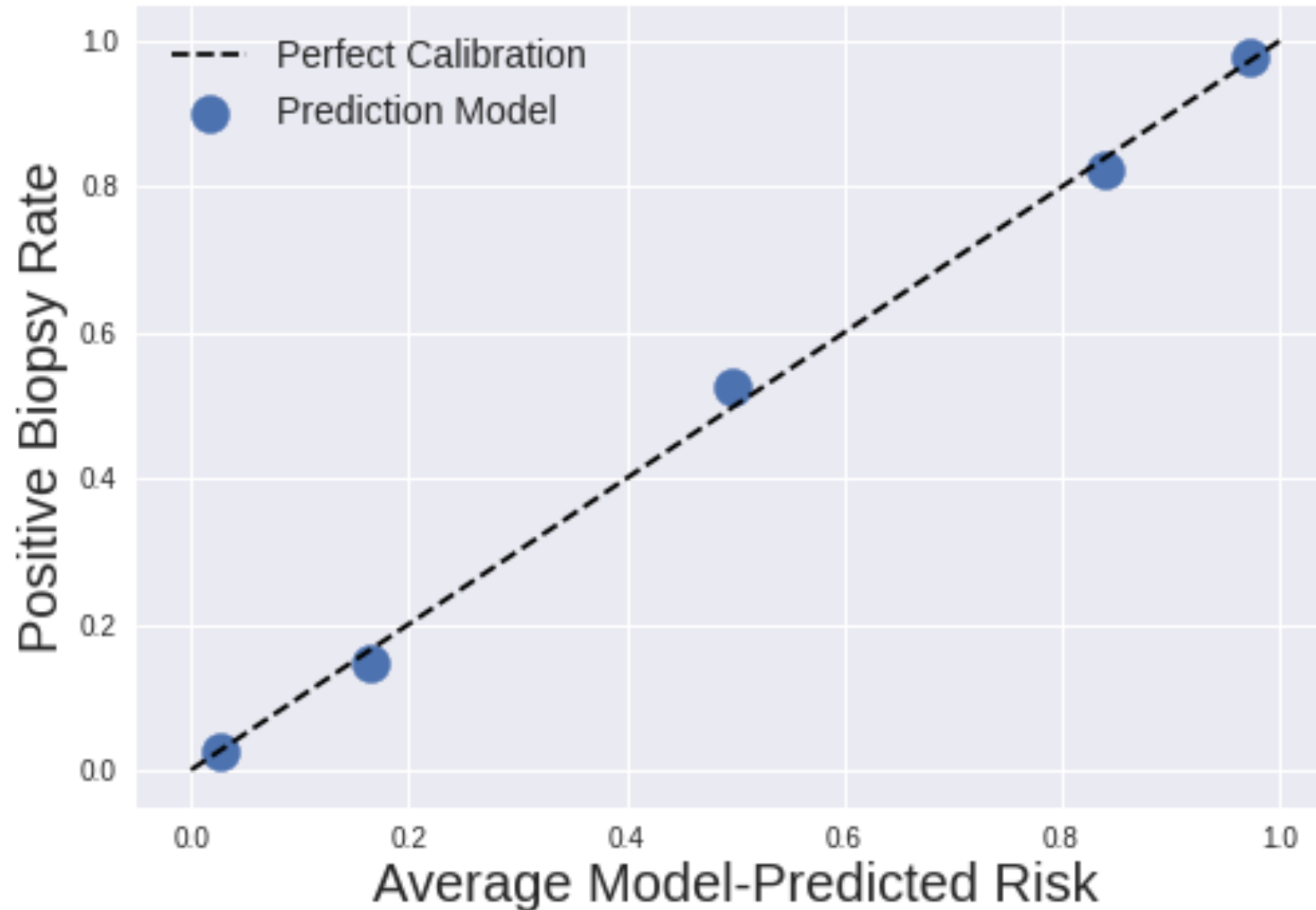
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5. Calibration: model calibration is equally good in both groups (?)

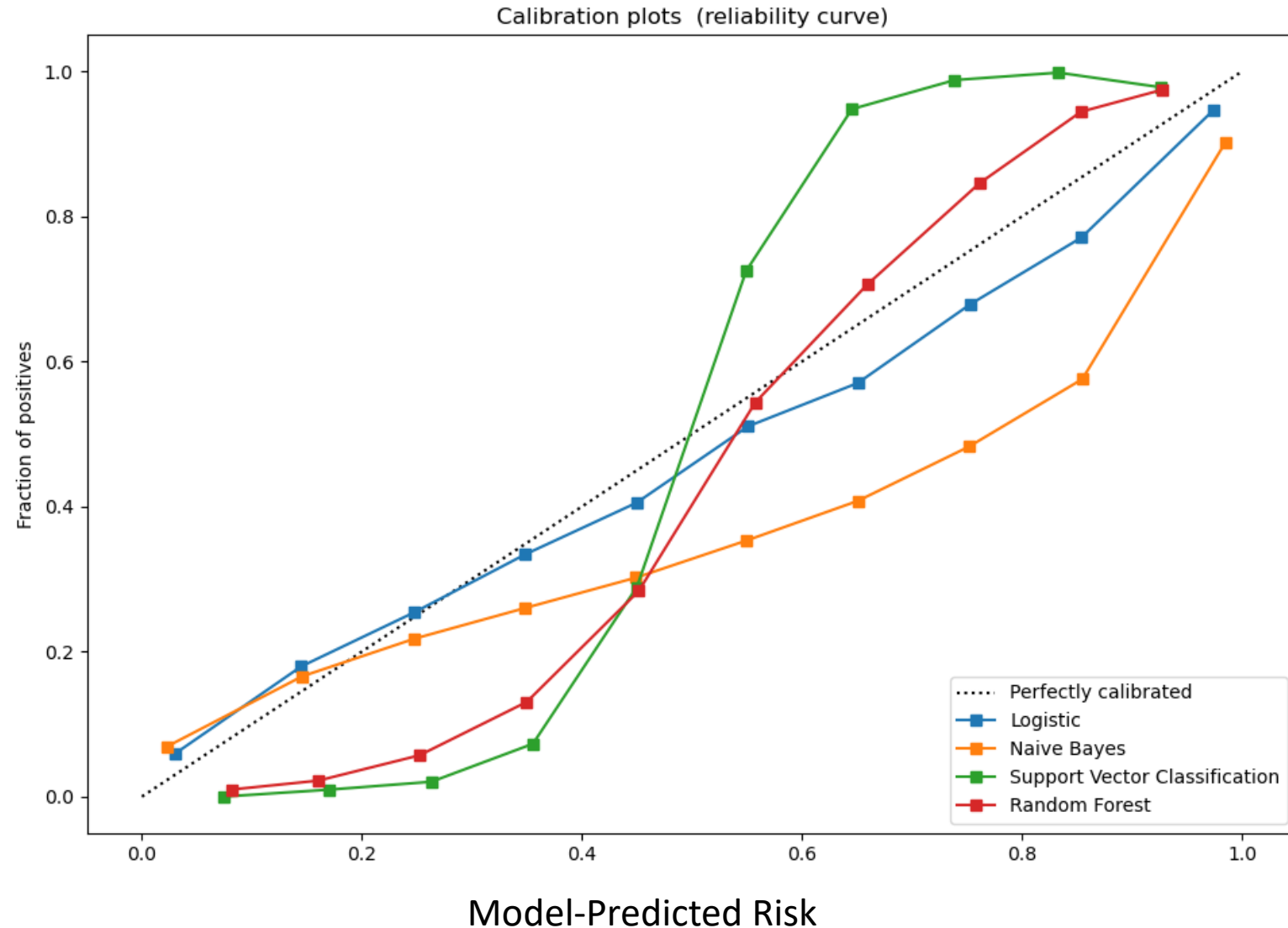
Except under special (& not realistic) circumstances, (1) – (4) are mutually exclusive.

Assessing calibration

Assess Calibration Graphically



Assess Calibration Graphically



Specific calibration metrics

- Quantile-based calibration slope and intercept
 - Associated Hosmer-Lemeshow test statistic
 - Limitation: sensitive to bin width
- Expected calibration error (ECE) 
 - Also sensitive to bin width
- Regression-based calibration slope and intercept
 - i.e. calibration in the large, calibration slope
 - Often the favored approach

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} \left| \text{acc}(B_m) - \text{conf}(B_m) \right|$$

where $B_1, \dots, B_M$ partition the n predictions into M bins (often by predicted confidence), and

$$\text{acc}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \mathbf{1}\{\hat{y}_i = y_i\}, \quad \text{conf}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \hat{p}_i.$$

Here $\hat{p}_i$ is the predicted confidence (e.g., $\max_k \hat{p}_{ik}$ in multiclass, or $\hat{p}_i$ for binary), $\hat{y}_i = \arg \max_k \hat{p}_{ik}$, and y_i is the true label.

Class balance

In health AI, we rarely have *balanced* classes

Diagnosis prediction

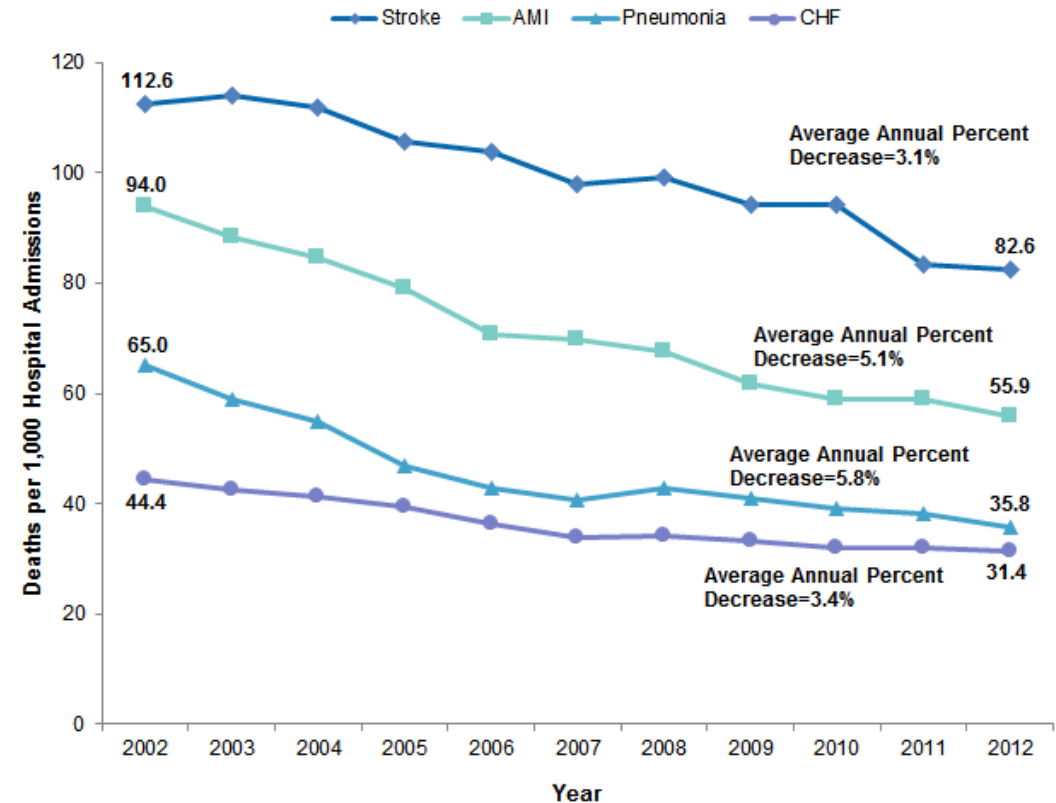
ADHD: 1 in 10

Autism: 1 in 31

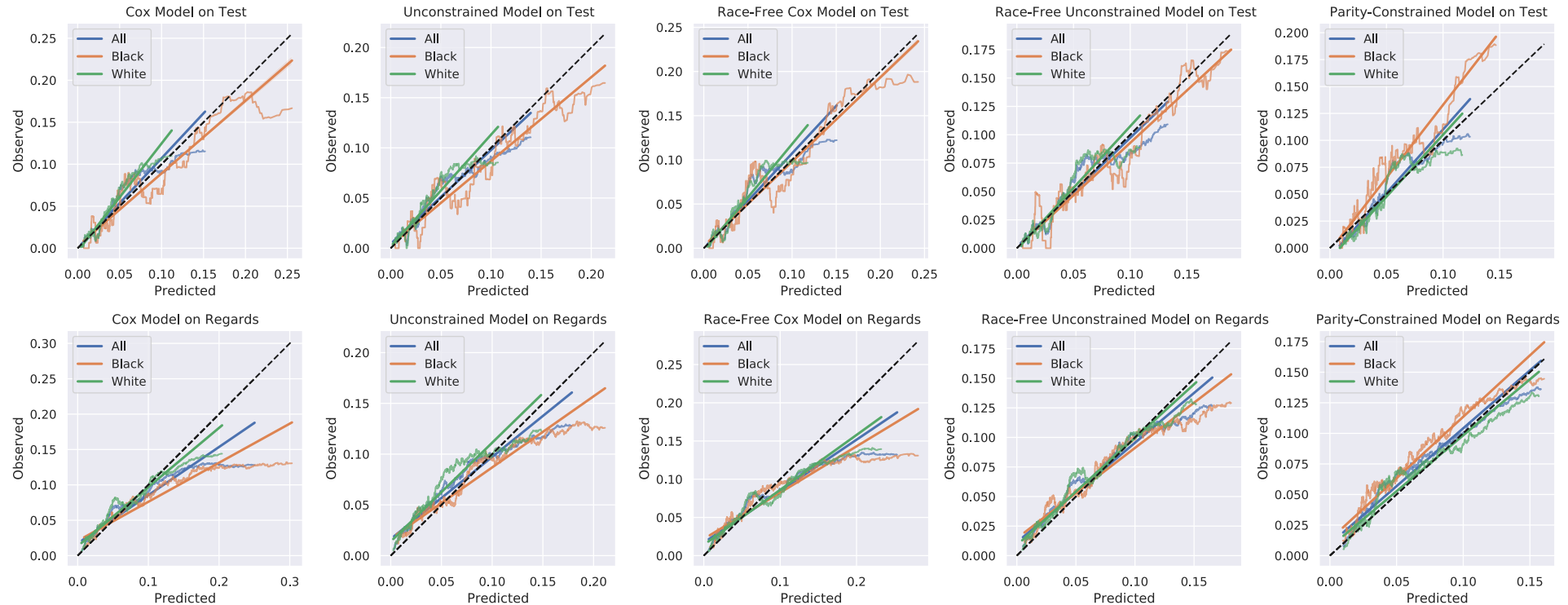
CF: 1 in 4000

Fibrodysplasia ossificans
progressiva (FOP): 1 in 1.5M

Mortality prediction



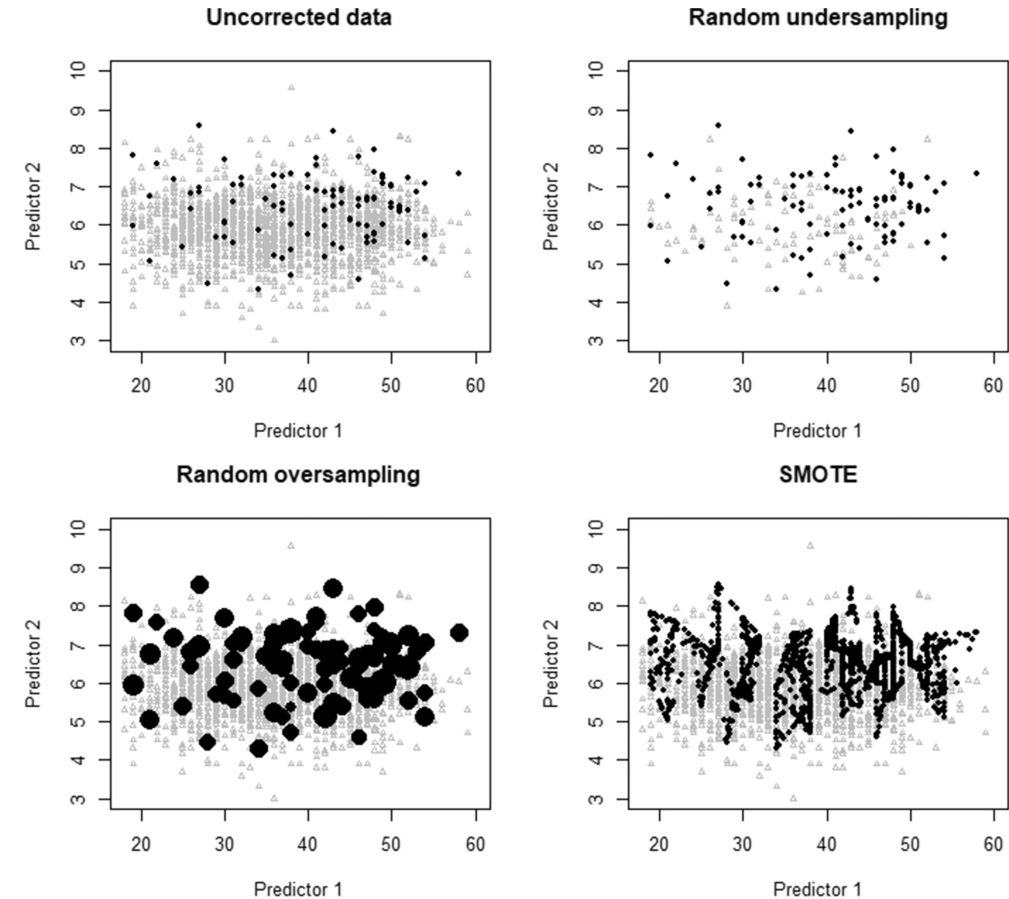
Assessing calibration for imbalanced problems



- Predicted probabilities tend to be close to 0
- Must use quantile-based bins
- `strategy='quantile'` when using `calibration_curve` in `sklearn.calibration`

Common techniques for imbalanced problems

- Over-/under-sampling
- Weighting the loss
- Synthetic data approaches, notably SMOTE
- What has been your experience with such techniques?
- Consider: gradient-based updates in mini-batches
- Consider: effects on calibration



Ruben van den Goorbergh, Maarten van Smeden, Dirk Timmerman, Ben Van Calster, The harm of class imbalance corrections for risk prediction models: illustration and simulation using logistic regression, *Journal of the American Medical Informatics Association*, Volume 29, Issue 9, September 2022, Pages 1525–1534, <https://doi.org/10.1093/jamia/ocac093>

Activity today: calibration and class balance

- Select a prediction task with imbalanced classes
- Train a model of your choice without correcting class imbalance
 - Assess AUROC
 - Assess calibration
- Correct class imbalance, then train again
 - Assess AUROC
 - Assess calibration
- Two groups will share results

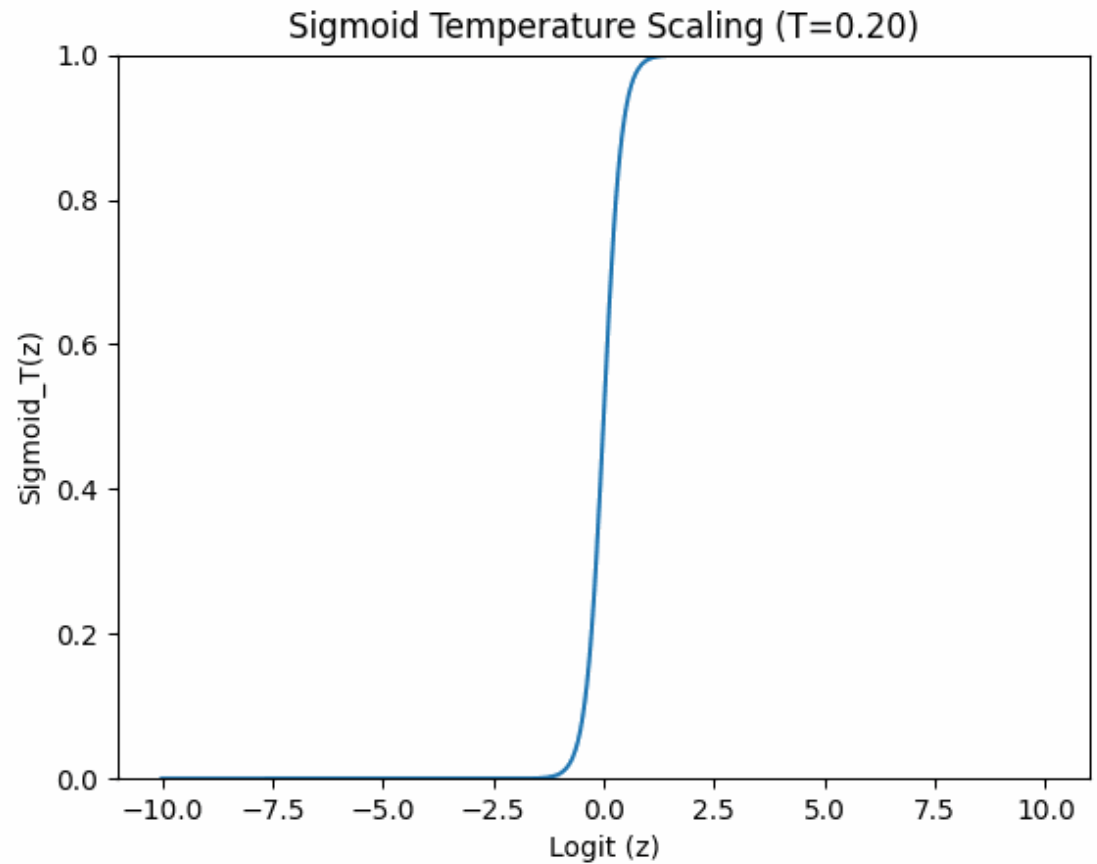
Post-hoc calibration techniques

From simple to not so simple

Temperature scaling: binary case

- In the typical case, our model is overconfident, often due to overfitting
- We aim to reduce this by reducing the temperature T of the sigmoid function
- Requires a calibration set

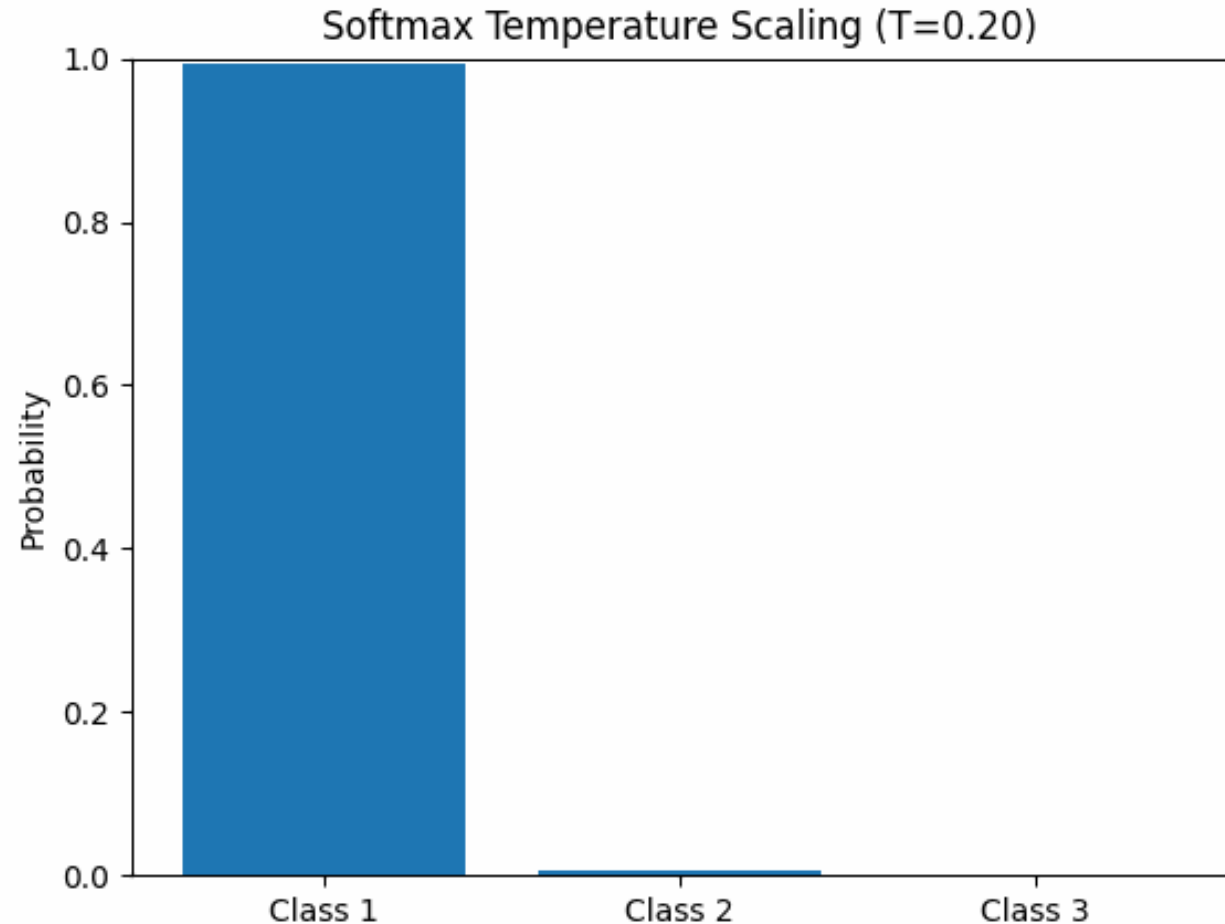
$$\sigma(z/T)$$



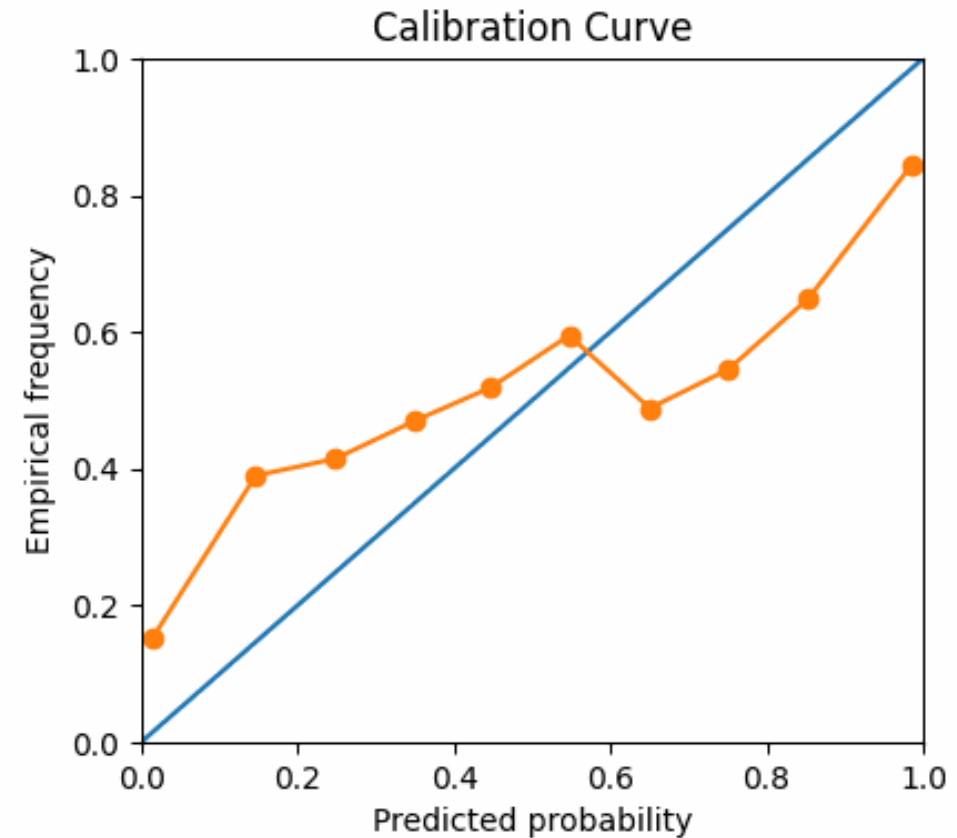
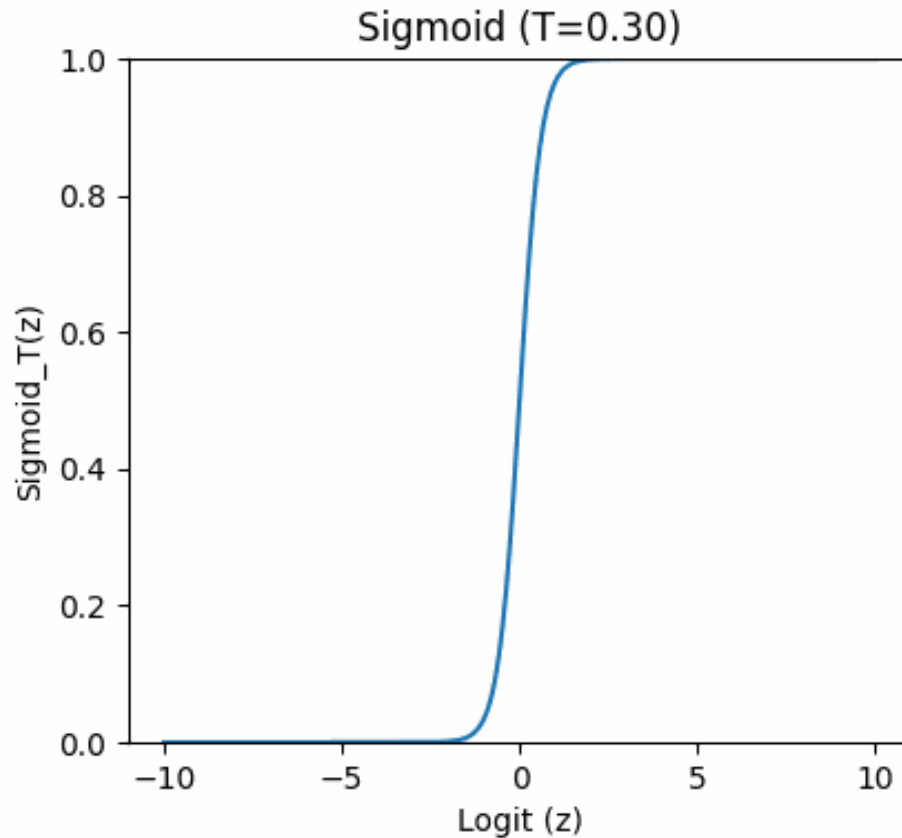
Temperature scaling: multi-class case

- In the typical case, our model is overconfident, often due to overfitting
- We aim to reduce this by reducing the temperature T of the sigmoid function
- Requires a calibration set

$$\text{softmax}(\mathbf{z}/T)$$

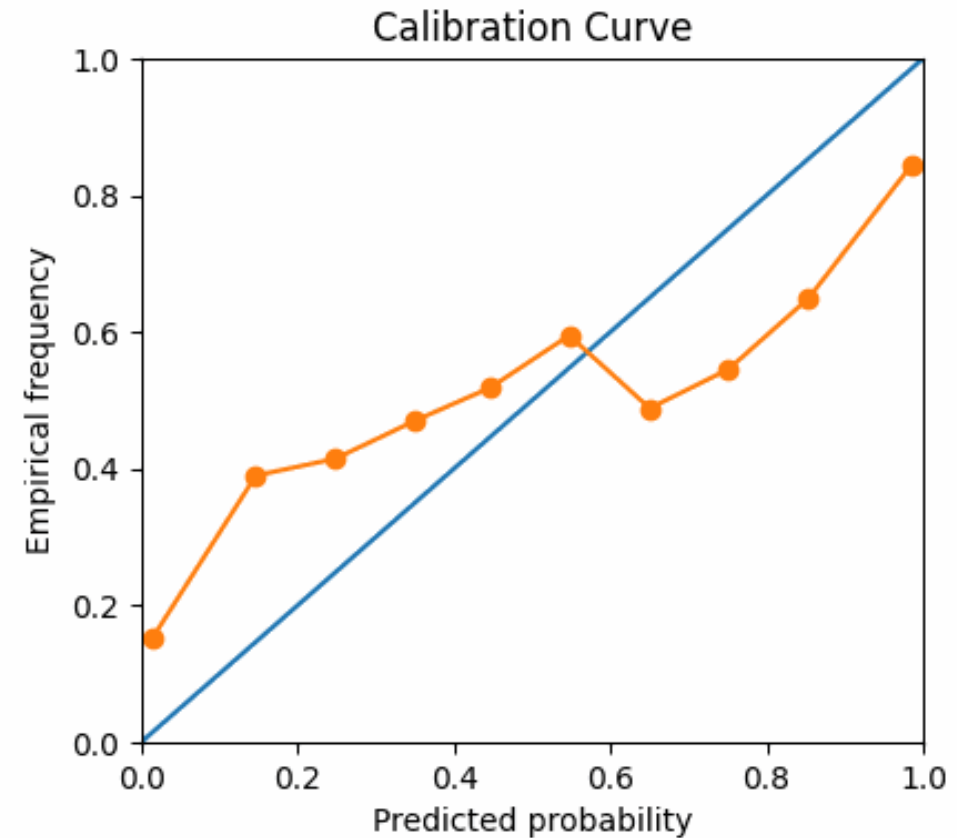
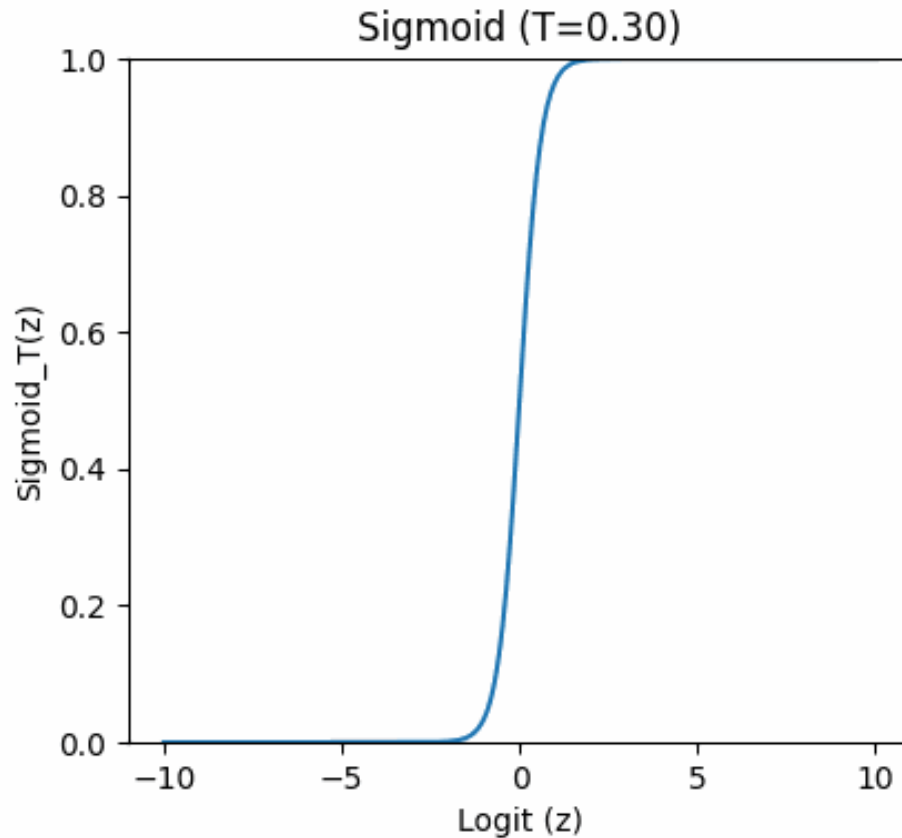


Temperature scaling: binary case



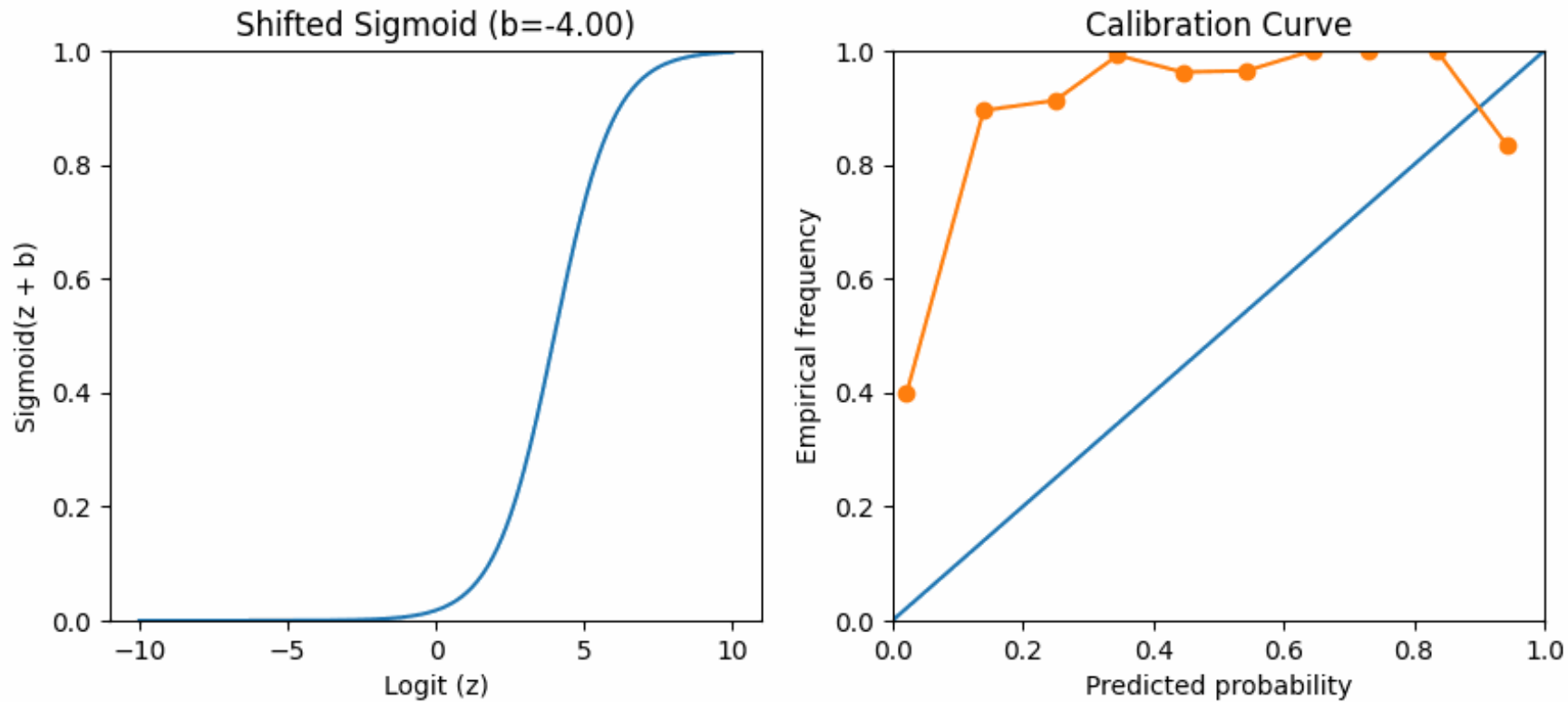
- Overconfident (probabilities are too extreme) -> calibrated -> underconfident (probabilities are not extreme enough)
- Reflected in the calibration slope

Temperature scaling: binary case



- Overconfident (probabilities are too extreme) -> calibrated -> underconfident (probabilities are not extreme enough)
- Consider the effects of parameter shrinkage (e.g., L2 or L1 regularization)

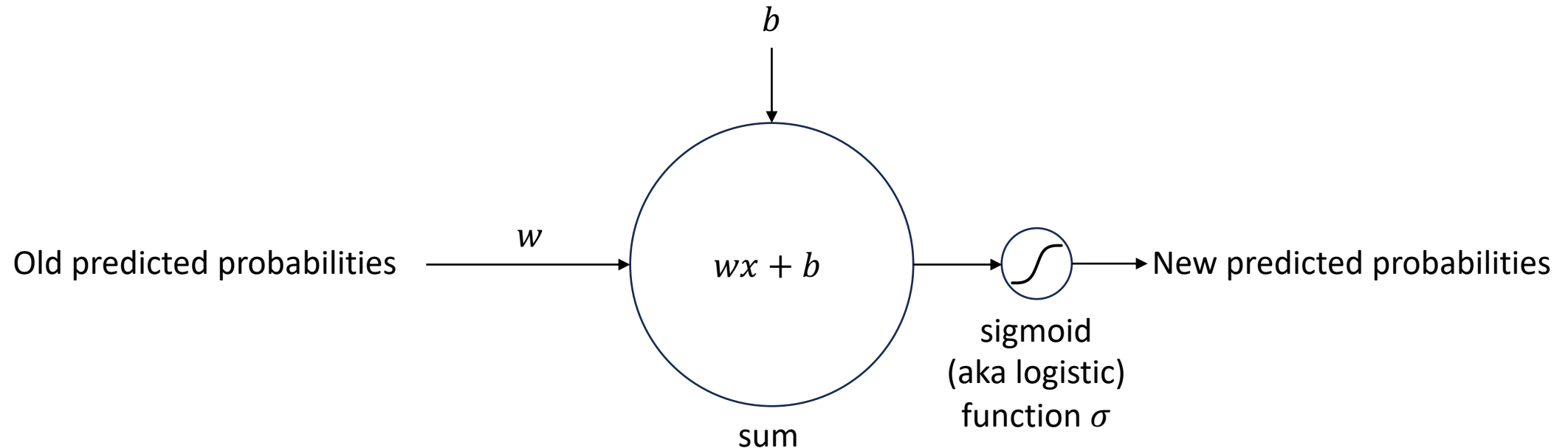
Temperature scaling: binary case



- Predicted probabilities too low $\rightarrow$ calibrated $\rightarrow$ predicted probabilities too high
- Reflected in the calibration in the large

Platt scaling

- Fit a logistic regression model on your predicted probabilities



- Or, a more flexible alternative: isotonic regression

Vector scaling

- $\text{softmax}(W\mathbf{z} + \mathbf{b})$
- Necessary when you need to rebalance for multi-class problems
- Requires a substantial calibration set

Calibration, discrimination, and
fairness

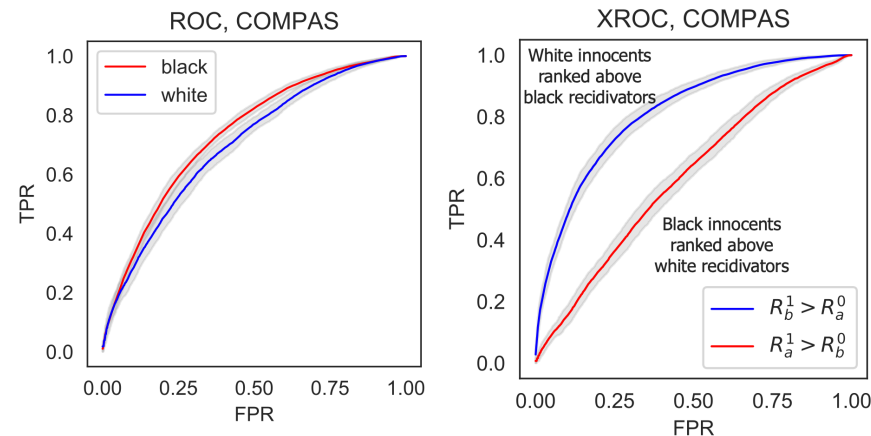
Assessing fairness with group conditional AUC (xAUC)

$$\text{xAUC}_{A,B} = P(S_+^A > S_-^B)$$

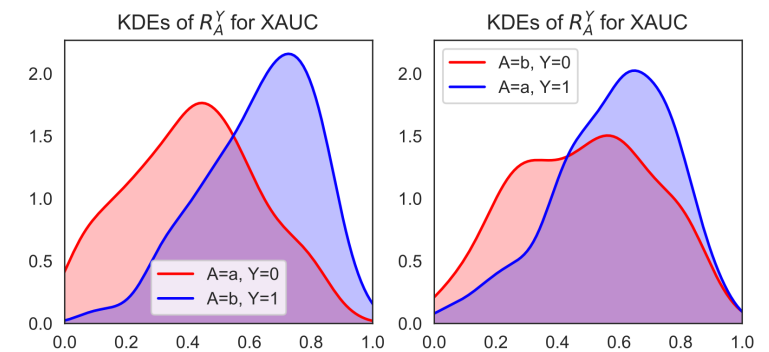
[i.e. we are now conditioning on group status in addition to pos/neg]

Consider $\Delta\text{xAUC} = |\text{xAUC}_{B,A} - \text{xAUC}_{A,B}|$

Does lower ΔxAUC
imply a fairer model?



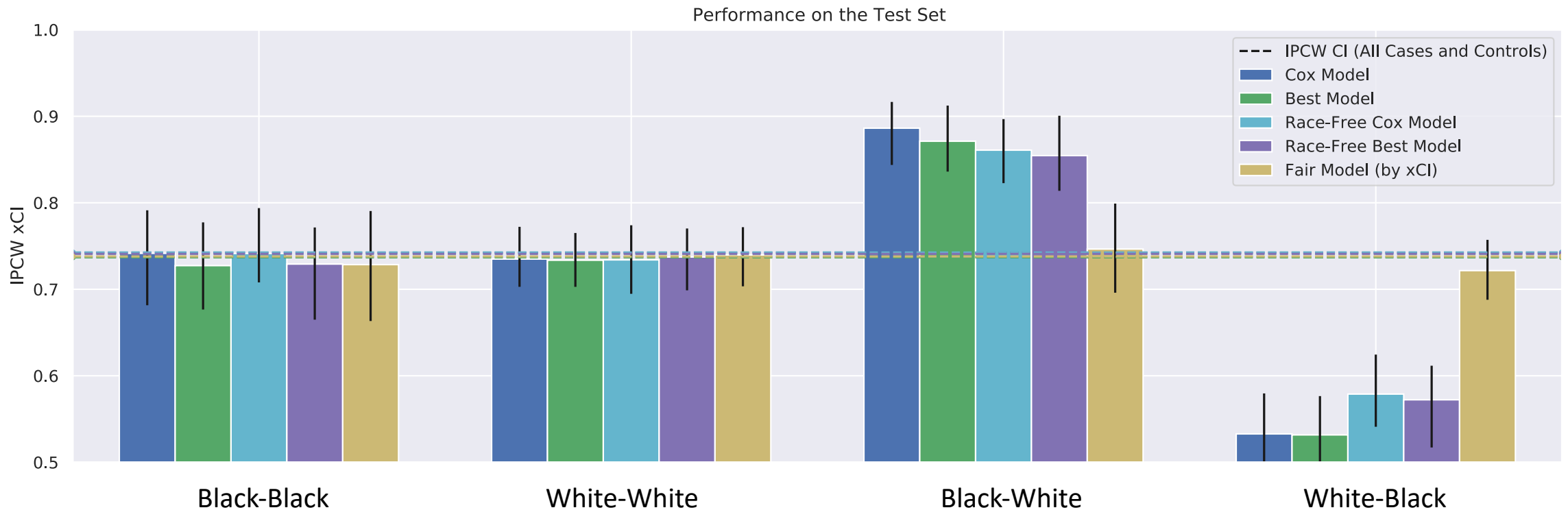
(a) ROC curves and XROC curves for COMPAS



(b) Score distributions $\Pr[R = r \mid A = a, Y = y]$

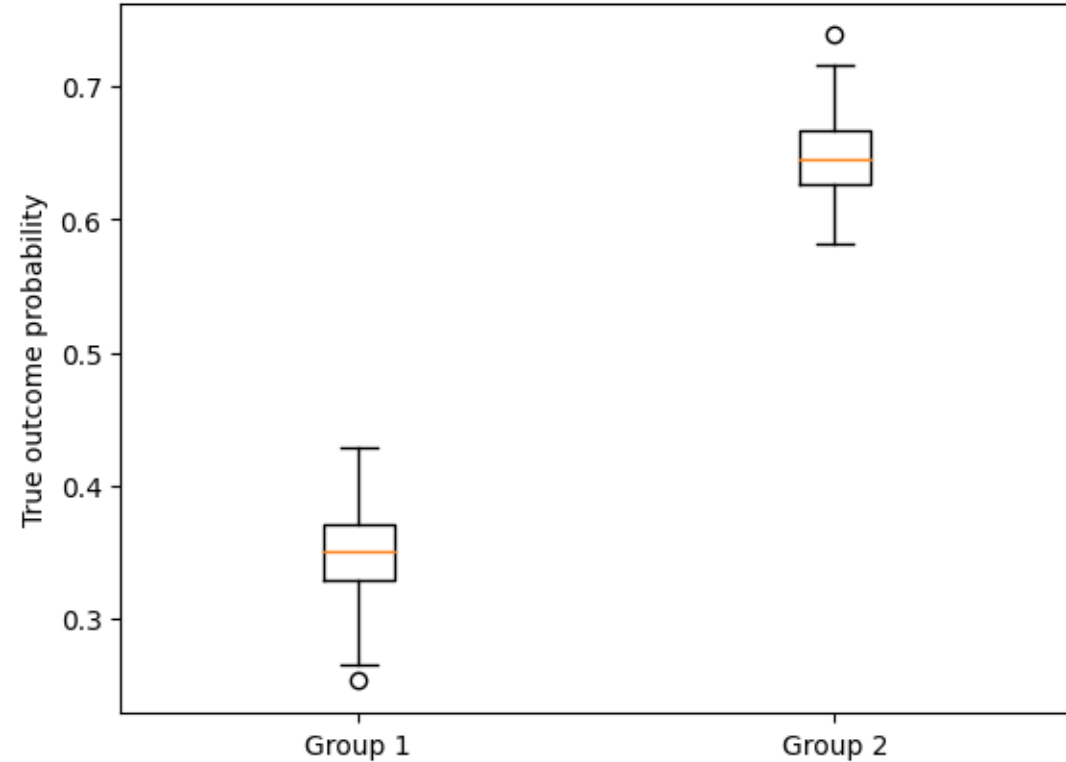
Our “fair” model is one that aims to minimize differences in xCI (similar to xAUC)

- Depending on the use case, this can be good or bad, fair or unfair.



Consider the xAUC for a calibrated model

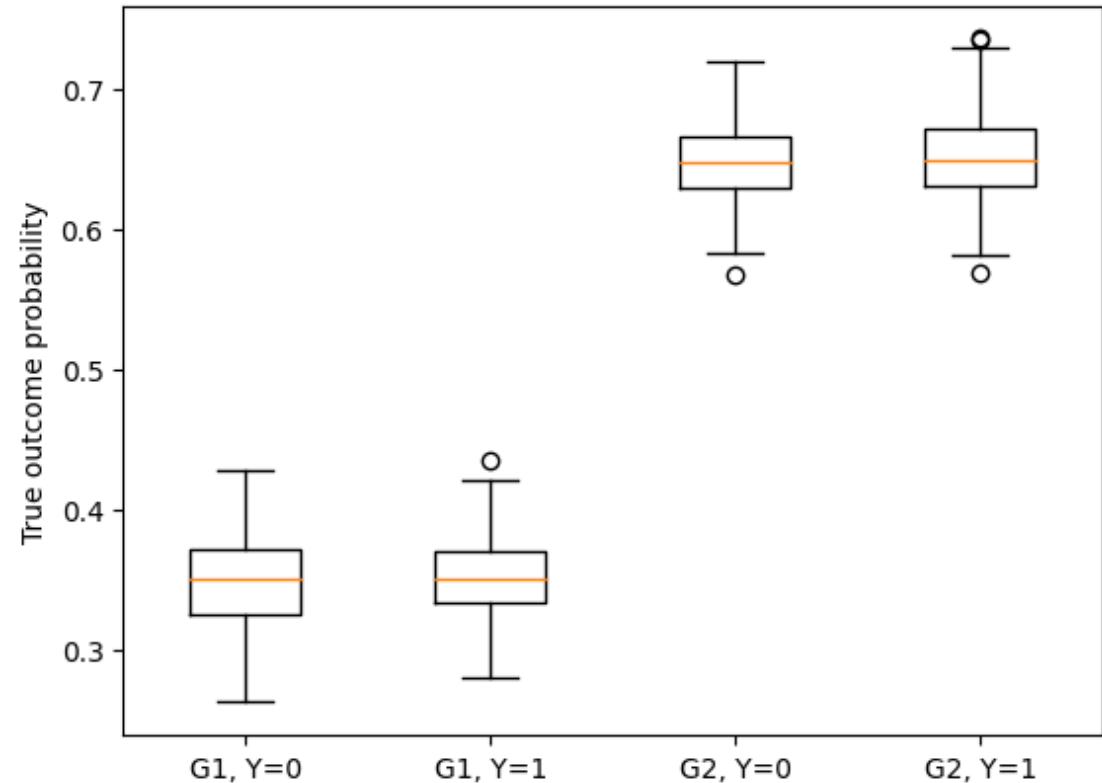
- Suppose there is a true difference between groups 1 and 2
- Suppose further that our model is optimal in the sense that its predictions are equal to the true outcome probabilities
- What is the $\text{xAUC}_{1,2}$?
- What is the $\text{xAUC}_{2,1}$?



$$\text{xAUC}_{1,2} = P(S_+^1 > S_-^2)$$

Consider the xAUC for a calibrated model

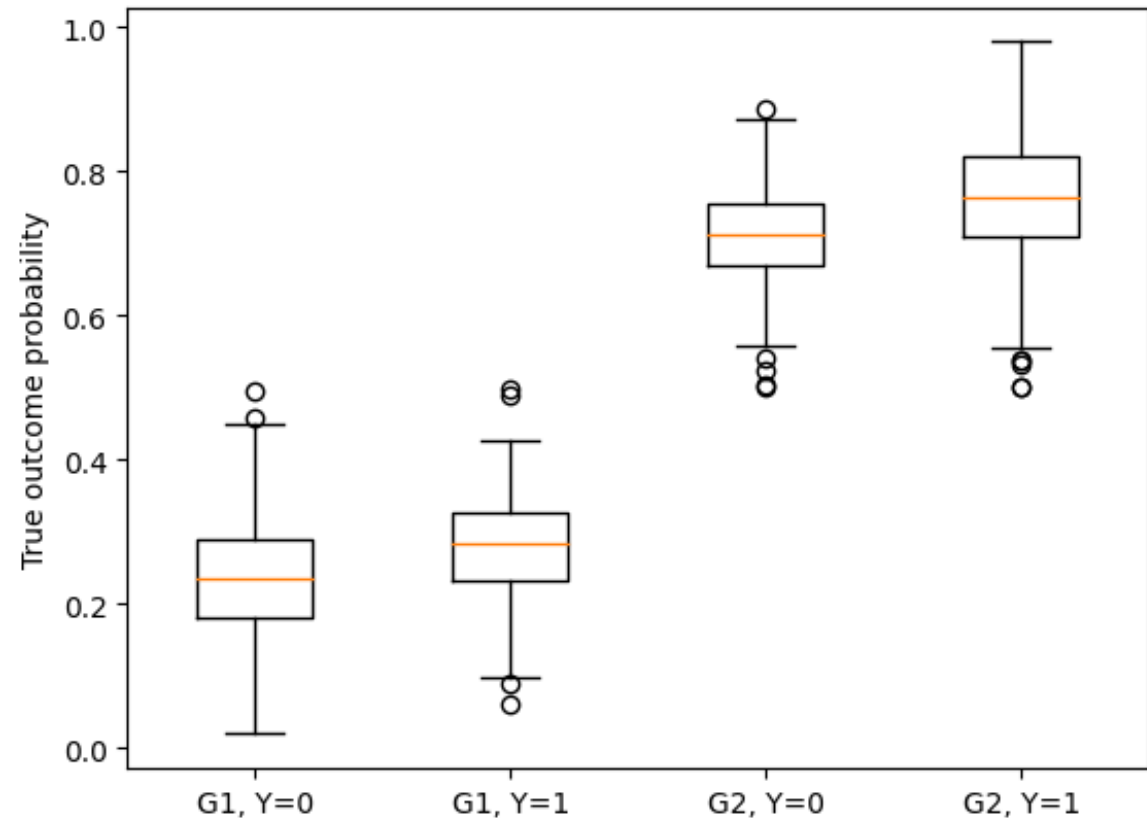
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- What is the $\text{xAUC}_{2,1}$?



$$\text{xAUC}_{1,2} = P(S_+^1 > S_-^2)$$

Consider the xAUC for a calibrated model

- Suppose there is a true difference between groups 1 and 2
- Suppose further that our model is optimal in the sense that its predictions are equal to the true outcome probabilities
- Are the distributions the same across groups after conditioning on outcomes? Why or why not?



$$\text{xAUC}_{1,2} = P(S_+^1 > S_-^2)$$

Report back

What constitutes a *fair* model?

1. A model with equal error rates between groups (and $\Delta x\text{AUC}=0$)
2. A model that is well-calibrated in both groups

Supposing a group difference in risk exists, these two criteria are mutually exclusive.